Rhythmic replay of short-term memory neural patterns revealed by time-resolved error prediction

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Abstract

Theta oscillations are thought to provide a temporal scaffold for short-term memory (STM), organizing item encoding and maintenance into successive phases to reduce representational conflict. Whether this rhythm also determines encoding fidelity, that is, how precisely items are encoded in human cortical activity, remains unclear. Here, we show that STM encoding fidelity fluctuates rhythmically at theta frequency. EEG was recorded while participants encoded arrays of colored, oriented objects under two memory loads and, after a delay, reported the features of a retrospectively cued item on a continuous scale. Using time-resolved multivariate pattern analysis, we predicted subsequent recall error from encoding- and maintenance-period activity. Fronto-parietal theta- and beta-band activity predicted subsequent memory error. This prediction was not sustained but fluctuated at theta frequency and was not modulated by memory load. Cross-temporal generalization indicated that the same neural pattern recurred across encoding and maintenance, underlying the rhythmic fluctuations in memory-error prediction. Prediction fluctuations were temporally offset across spatial positions and object features. These findings characterize STM encoding as a rhythmic, recurrent process and link behavioral theta fluctuations to a distributed neural mechanism.

Keywords: *Short-term memory; theta oscillations; EEG; multivariate pattern analysis; memory encoding; memory error prediction*

Introduction

Short-term memory (STM) is the capacity to temporarily store and process a limited amount of information for ongoing cognitive purposes. Two central processes are important for this: encoding, which transforms a transient, multi-feature sensory scene into memory representations, and maintenance, which preserves these representations across a delay. One long-standing question is how the brain coordinates the encoding of several items, each defined by multiple features, into stable yet separable traces.

A substantial body of evidence links memory to neural oscillations, particularly theta-band activity and higher-frequency oscillations in the beta and gamma ranges (Lisman & Jensen, 2013; Lundqvist et al., 2016). Theta power has been repeatedly associated with successful encoding, with stronger theta activity for subsequently remembered than forgotten items (Sederberg et al., 2003; Hanslmayr & Staudigl, 2014). Delay-period theta increases have further been linked to the maintenance of task-relevant information (Jensen et al., 2002; Hsieh & Ranganath, 2014).

Neurophysiologically, theta is hypothesized to coordinate communication across distributed cortical and hippocampal regions and to govern the timing of faster oscillations, such that memory function relies not on a single locus but on a distributed network with multi-frequency interactions (Sauseng et al., 2010; Staresina & Wimber, 2019). Beyond the mechanistic, spatiotemporal organization of memory network interactions, theta is also proposed to organize *how* mnemonic information is structured in time — how successive items are coded, how concurrent representations are kept apart, and how encoding and retrieval are temporally separated. Within a theta cycle, individual memoranda are reactivated in sequence, each carried by a distinct gamma or beta subcycle (Lisman & Idiart, 1995; Jensen & Tesche, 2002; Heusser et al., 2016), so that theta phase segregates concurrently held items and replays them with reduced mutual interference.

The same phase code is argued to address a more basic computational demand: distinguishing information arriving from the external sensory stream from information already encoded and internally regenerated. Theta phase is proposed to group mnemonic content such that the neural operations supporting encoding and retrieval occupy separate, opposing phases of the cycle (Hasselmo et al., 2002; Kerrén et al., 2018), providing one phase favorable to registering new input and another favorable to retrieval and reinstatement during maintenance. Critically, encoding and maintenance are not cleanly successive stages: maintenance begins as material is encoded, and theta-phase organization of individual items is already evident during encoding itself (Han et al., 2026). On this view, the encoding of a multi-item scene and the maintenance of its trace are interleaved within an ongoing theta rhythm rather than separated into discrete epochs.

Direct evidence for theta-phase modulation of the interaction between encoding and replay of short-term memory representations in humans remains limited. Functional MRI cannot, by its nature, resolve the rapid alternation of cortical patterns underlying memory formation and reinstatement. Behavioral studies show that the moment participants report a successful recollection is itself locked to a particular theta phase (Ter Wal et al., 2021; Abdalaziz et al., 2023), and time-resolved MVPA of EEG and MEG indicates that memory content can be decoded from distributed cortical patterns rhythmically: classification accuracy for stored content — for instance, indoor versus outdoor scenes, or animate versus inanimate objects — fluctuates at theta frequency and peaks at a specific phase of frontal and hippocampal theta (Fuentemilla et al., 2010; Kerrén et al., 2018). Three questions, however, are left open by this work. First, it is unknown whether these decoding fluctuations track the *fidelity* of memory or merely reflect the periodic reinstatement of sensory content, irrespective of how well it is retained. Second, it is unclear whether the fluctuations index a sequential sweep through the visual scene — successive, distinct neural patterns processing different content — or the genuine recurrence of a single reinstated pattern; distinguishing these requires testing whether the recurring code is the same across time, which cross-temporal generalization can address

(King & Dehaene, 2014). Third, because previous MVPA studies classified whole scenes, it remains unknown whether one theta-rhythmic process governs an entire multi-item array or whether separable rhythmic dynamics can be resolved for individual items and for the distinct features composing them.

In the present study, we used time-resolved MVPA to predict continuous memory error directly from encoding and maintenance period EEG, addressing the three questions raised above. Participants encoded arrays of objects defined by color and orientation under two memory loads (two or four spatially distanced items) and, after a delay, reported a cued feature of a cued item on a continuous scale. Rather than classifying which stimulus was held in memory, we trained predictors to estimate the magnitude of subsequent recall error for the color and orientation of individual memoranda, and asked when, and through which oscillatory activity, the encoding pattern carried this information. This shift from decoding content to predicting fidelity speaks to the first question: whether the rhythmic decoding reported previously tracks how well an item is memorized or merely the periodic reinstatement of its sensory content. To address the second question — whether recurring prediction reflects a sequential sweep of distinct neural codes or the recurrence of a single code — we applied cross-temporal generalization (King & Dehaene, 2014), testing whether a predictor trained at one moment of encoding generalizes to later moments. To address the third, we resolved prediction separately for each item and for its two features, allowing us to ask whether one rhythmic process governs the whole array or whether separable dynamics operate at the level of individual items and features.

We found that fronto-parietal theta- and beta-band activity during encoding predicted subsequent memory error. Rather than being sustained, item-specific prediction fluctuated at a theta rate (~3–5 Hz), with the same neural pattern recurring across encoding and into maintenance, as shown by cross-temporal generalization. Prediction was temporally offset across spatial positions, and color and orientation followed dissociable dynamics, suggesting that item features were encoded through distinct temporal processes. Together, these findings support a rhythmic, recurrent organization of multi-item STM encoding and have implications for computational models of STM and temporally targeted neuromodulation of memory processes.

Methods

Participants

Twenty-one participants (11 female; 10 male, mean age 25, range 19–34) took part in this study. All, except one participant, were right-handed, English speaking and had normal or corrected-to-normal vision. The study was approved by the Ruhr University Bochum Research Ethics Committee (Protocol number: 2025-416-f-S). All participants provided written informed consent prior to participation and were informed about the experimental procedure.

Short-term memory task design

Task and stimuli. Participants performed a delayed continuous-report visual short-term memory task implemented in PsychoPy (Peirce, 2007). Each trial began with a fixation cross of jittered duration (800–1300 ms), followed by the encoding period, during which the encoding array was presented. The array consisted of either 2 (low load) or 4 (high load) colored, oriented rectangles, each presented at one of five fixed locations (upper-left, upper-right, lower-left, lower-right, and center). The occupied locations varied randomly from trial to trial, so that participants could not anticipate at which positions items would appear. For each trial, item colors and orientations were constrained to be mutually distinct. Colors were sampled from a circular CIELAB hue space (L = 60, chroma = 38) with a minimum pairwise separation of ~79°. Orientations were selected from a discrete set ranging from 0° to 170° in 10° increments, with a minimum pairwise separation of 30°.

The encoding period lasted 800 ms and was followed by a 700 ms maintenance interval, a 300 ms visual mask (to remove iconic memory effects ; Coltheart, 1983), and a subsequent 1000 ms delay.

After the delay, a spatial retro-cue was presented for 2000 ms, indicating which of the encoded items was to be reported. Because the cued item was unpredictable, participants were required to maintain all items. Following the cue, participants reported the remembered features of the cued item using continuous responses: color was selected on a continuous color wheel with the computer mouse, and orientation was reproduced by rotating a rectangle using the mouse wheel. Response time was unrestricted. The order of color and orientation reports was randomized across trials, minimizing potential order effects on memory fidelity.

Design and procedure. Each block contained 40 trials, with 10-second breaks after every 10 trials. The session comprised four blocks: three high-load (4 items) and one low-load (2 items). This resulted in a total of 160 trials (120 high-load and 40 low-load). The order of the blocks was randomized between subjects. Before the main task, participants received instructions, were asked to minimize blinking during the encoding and maintenance phase, and completed a 5-trial practice run to confirm task understanding. Including EEG cap preparation, each session lasted no more than 2 hours.

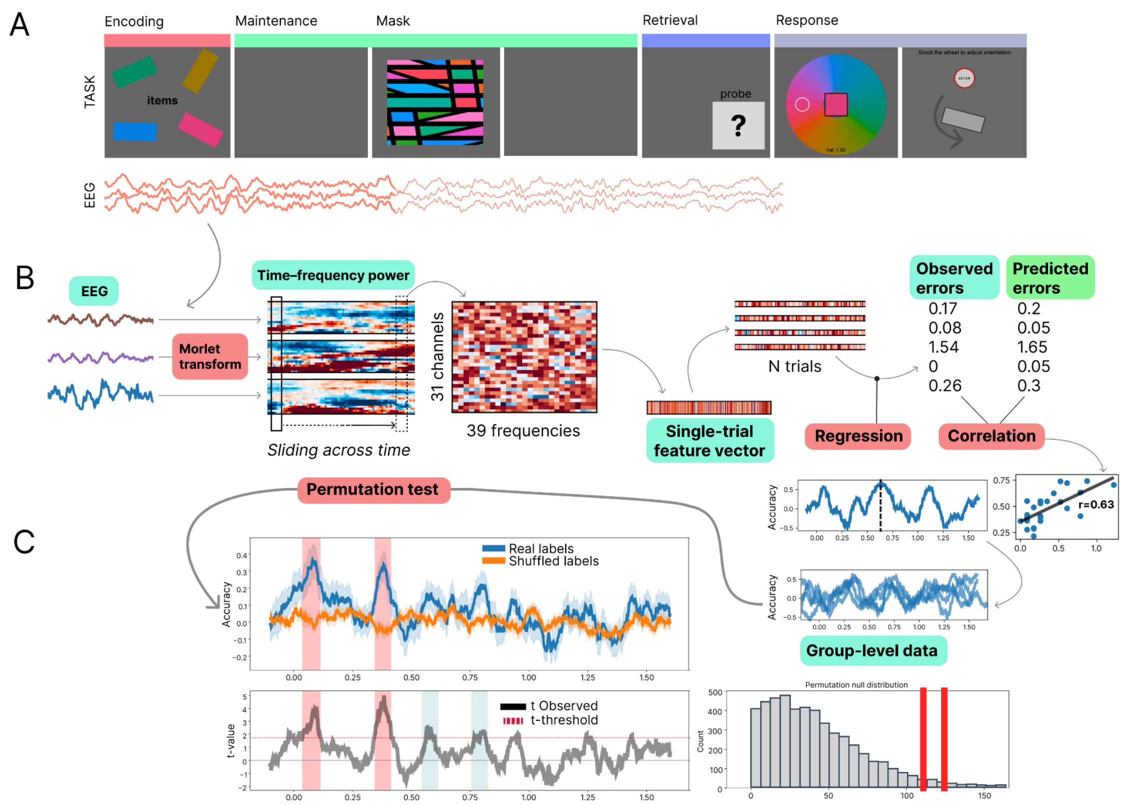

Figure 1. Delayed continuous-report visual short-term memory task and multivariate decoding analysis pipeline.

A. Delayed continuous-report STM task. Participants encoded two or four colored, oriented rectangles presented at fixed spatial locations, maintained them across a delay, and reported the color and orientation of a retrospectively cued item using continuous-response measures.

B. Time-resolved decoding analysis. EEG signals with evoked activity subtracted were transformed into time–frequency power using Morlet wavelets. At each time point, channel × frequency power values were concatenated into single-trial feature vectors and entered a ridge-regression model to predict trial-by-trial memory error separately for color and orientation.

C. Group-level statistical analysis. Decoding accuracy was quantified as the correlation between observed and predicted errors and evaluated over time. Statistical significance was assessed using a one-sided cluster-based permutation test, comparing observed decoding performance against a surrogate distribution obtained from shuffled error labels.

Behavioral data analysis

Color responses and targets were expressed in radians on a circular hue wheel (full wheel = 2π), whereas orientation responses and targets were expressed in radians within a 0–π range. Color errors were computed as signed circular distances wrapped to (−π, π], and orientation errors as signed minimal distances in 180°-periodic space, wrapped to $(-\pi/2, \pi/2]$. Absolute errors were defined as the magnitudes of these signed errors.

To assess dependence between feature-specific errors, for each participant, mutual information (MI) between absolute color and orientation errors was estimated using a k-nearest-neighbor estimator. The group statistic was defined as the mean MI across participants. Significance was assessed against a permutation null distribution generated by shuffling errors within each participant (1,000 permutations) and recomputing the group mean. Analyses were performed separately for each memory load.

To test whether recall error varied with target location, feature, and memory load, absolute errors were first normalized by the maximum possible error for each feature, accounting for their different circular ranges (0–π for color; 0–$\pi/2$ for orientation). Normalized absolute errors were averaged within participant, feature, and location and entered into a repeated-measures ANOVA with Feature (color, orientation), Location (five levels), and Memory load (2-, 4-items) as within-participant factors. Greenhouse–Geisser correction was applied to factors with more than two levels.

Mixture-model decomposition. To estimate the sources of memory error, we fit the mixture model of Bays et al. (2009) — decomposing continuous-report errors into target, swap (misbinding), and guess components — separately to color and orientation responses using the biased_memory_toolbox (Zhou et al., 2022). Participant-specific parameters were then used to derive trial-wise posterior probabilities of target, swap, and guess responses (P_hit, P_swap, P_guess) via Bayes' rule.

EEG recording

Participants were seated comfortably in front of a computer screen and completed practice trials to familiarize themselves with the task. EEG was recorded using an ActiChamp amplifier (Brain Products, Germany) at a sampling rate of 1 kHz from 31 scalp electrodes positioned according to 10–10 system at Fp1, Fp2, F7, F3, F1, Fz, F2, F4, F8, FC5, FC3, FC1, FCz, FC2, FC4, FC6, C3, Cz, C4, T7, T8, CP5, CPz, CP6, P7, P3, Pz, P4, P8, O1, and O2. Channel TP10 was used as the recording reference, and the data were subsequently re-referenced to the common average. Electrode impedances were maintained below 10 kΩ for all channels. EEG acquisition was synchronized with stimulus presentation using an analog photosensor attached to the screen and connected to a Micro1404 interface (Cambridge Electronic Design, UK), which was in turn linked to the EEG amplifier.

EEG preprocessing

Preprocessing was performed in MNE-Python (Gramfort et al., 2013). Continuous EEG data were first notch-filtered at 50 Hz and its harmonics (100, 150, and 200 Hz) to suppress line noise and subsequently band-pass filtered between 0.1 and 70 Hz using a zero-phase FIR filter. Noisy channels were identified by visual inspection and repaired using spherical spline interpolation (mean = 0.38 channels per participant), after which the data were re-referenced to the common average. Independent component analysis (FastICA) was then performed to remove ocular and muscle artifacts. To improve component decomposition, ICA was fitted to a copy of the data band-pass filtered between 1 and 40 Hz and retaining 99% of the variance, and the resulting decomposition was subsequently applied to the broadband (0.1–70 Hz) data. Components corresponding to eye-movement and muscle artifacts were identified by visual inspection and removed.

Continuous data were segmented into epochs from −1 to +2.5 s relative to memory-array onset, covering the encoding and maintenance intervals. Epochs were baseline-corrected using the −200 to 0 ms pre-stimulus window and downsampled to 500 Hz.

Bad epochs were detected and corrected using AutoReject (Jas et al., 2017), which repairs or rejects epochs based on cross-validated, channel-specific peak-to-peak amplitude thresholds (consensus κ and the number of interpolated channels ρ optimized by cross-validation over $\kappa \in [0.5, 1.0]$ and $\rho \in \{0, 1, 2, 3\}$). Channels that exceeded their threshold within an epoch were flagged. Epochs were rejected when the proportion of flagged channels exceeded the consensus criterion (κ), and otherwise retained with up to three affected channels interpolated. On average, 2 epochs (1.25%) were rejected per participant.

Time-Frequency Decomposition and Induced Spectral Power

Induced (non-phase-locked) spectral power reflecting oscillatory activity without contribution of evoked response was obtained (Luu et al., 2004; Trujillo & Allen, 2007). This was done with a single-trial subtraction approach via removing the evoked response directly from each trial's complex spectral representation prior to power computation (Kalcher & Pfurtscheller, 1995). Time-frequency spectral power dynamics was then performed using complex Morlet wavelet transform. The frequencies of the wavelets ranged from 3 Hz to 40 Hz with 1 Hz step. The full-width at half-maximum (FWHM) was 187 ms.

Multivariate pattern analysis

Time-resolved multivariate pattern analysis (MVPA) was used to predict trial-wise recall error from the spatial–spectral distribution of oscillatory power. Decoding was performed separately for color and orientation errors and, within each feature dimension, separately for each cued spatial position. Because two or four items were presented simultaneously at fixed locations and only one item was retrospectively cued for report, position-specific decoding was motivated by the hypothesis that items at different locations are sampled with distinct temporal dynamics during encoding.

For each participant and time point, feature vectors were constructed by concatenating power values across channels and frequencies. Features were standardized within each cross-validation fold and entered into an L2-regularized linear regression model (ridge regression; $\alpha = 1.0$) trained to predict absolute STM error, defined as the magnitude of the wrapped circular distance between the reported and target feature value ($0–\pi$ for color; $0–\pi/2$ for orientation).

Model performance was evaluated using 5-fold cross-validation. Decoding accuracy was assessed as the Pearson correlation between predicted and observed errors across folds. Repeating this analysis at successive time points (step size = 1 sample) yielded a time-resolved decoding-accuracy time course. Surrogate accuracy time courses were generated by randomly permuting error values across trials and repeating the full cross-validated MVPA procedure. This was performed using five independent permutations per participant, spatial position, and feature dimension, and surrogate performance was averaged across permutations in the same manner as the observed data.

Cross-subject generalization. To test whether the neural patterns predictive of recall error were consistent across individuals, we used leave-one-subject-out (LOSO) cross-validation. At each time point, a ridge model was trained on the pooled power features of $N - 1$ subjects and used to predict the absolute recall error of the held-out subject, yielding one LOSO prediction timecourse per subject. Surrogate timecourses for the LOSO analysis were generated using the same error-shuffling procedure as the within-subject decoding.

Group-level statistics. Group-level significance was assessed using a one-sided cluster-based permutation test across time (5,000 permutations). Clusters were defined using a cluster-forming threshold $t = 1.725$ ($p < 0.05$, tail = 1; df = 20) and considered significant at the cluster level when $p < 0.05$.

Feature contribution analysis. To characterize which channels and frequencies were most contributing in color and orientation error prediction, we examined the coefficients used for MVPA models. For each participant, regression weights were extracted from the time windows showing significant group-level decoding. Within each model, weights were z-scored across features and averaged across the selected time windows and across target item positions (color and orientation were analyzed separately to test if there are feature-specific patterns). The resulting per-participant channel × frequency maps were tested against zero at the group level with a two-sided cluster-based permutation test (10000 permutations; cluster-forming threshold $t = 2.0$; MNE-Python spatio_temporal_cluster_1samp_test), using channel adjacency obtained from the montage via mne.channels.find_ch_adjacency so that clusters could extend across neighboring channels and adjacent frequency bins.

Cross-temporal generalization of error prediction. To assess the temporal stability of the neural patterns predicting recall error, we performed a cross-temporal generalization (CTG) analysis (King & Dehaene, 2014). Following the same MVPA pipeline and cross-validation procedure as the main time-resolved decoding analysis, a ridge model trained at each encoding time point was tested at every other time point, which resulted in a train × test generalization matrix, where each cell contains the Pearson correlation between predicted and observed error. Group-level matrices were computed separately for color and orientation and for each target spatial position and tested against zero with a one-sided cluster-based permutation test (10,000 permutations; tail = 1; cluster-forming threshold $t = 1.725$, $p < .05$ one-sided, $df = 20$). Significant prediction confined to the diagonal (train time = test time) would indicate a succession of transient, temporally specific encoding codes, whereas off-diagonal clusters separated from the diagonal would indicate reactivation of an earlier code; a broad, sustained off-diagonal (square) pattern would indicate a stable maintained code.

Temporal relationships between STM error predictions across spatial locations and features

Phase differences across spatial locations. To test whether the 4 Hz phase of the decoding rhythm varied with target location, a 4 Hz sinusoid was fitted to each participant's decoding time course separately for each of the five target coordinates, and the phase of the fitted oscillation was extracted. To remove inter-individual differences in absolute phase, each participant's coordinate phases were expressed relative to that participant's circular mean phase across coordinates. The centered phases were entered into a one-way Watson–Williams circular ANOVA with target coordinate as a five-level factor.

Color and orientation error prediction phase coupling analysis. To test whether the rhythmic fluctuations in color and orientation decoding were temporally coordinated, we assessed whether the two features maintained a consistent phase offset across participants, pooling spatial coordinates as repeated observations.

Each time course was linearly detrended (scipy.signal.detrend), mean-subtracted, and multiplied by a Tukey window ($\alpha = 0.5$). We extracted the 4 Hz phase of each decoding time course from its complex Fourier coefficient at 4 Hz, and took the color–orientation phase difference ($\Delta\varphi = \varphi_{color} - \varphi_{orientation}$) for each participant and coordinate.
Because different target item spatial coordinates within a participant are not independent, group-level consistency of $\Delta\varphi$ was tested with a subject-blocked permutation Rayleigh test (Rayleigh test for non-uniformity; Berens, 2009): the observed Rayleigh statistic over all subject–coordinate observations was compared with a null distribution (10,000 permutations) in which each participant's complete set of coordinate phase differences was rotated by a single random angle, preserving within-subject structure while removing group-level phase alignment. A significant result of this testing would mean $\Delta\varphi$ clustered around a consistent direction across participants rather than being uniformly distributed — i.e., color and orientation decoding held a reliable, non-random 4 Hz phase relationship at the group level

As a complementary test of whether $\Delta\varphi$ was organized by target location rather than reflecting a single global offset, we ran a within-subject condition-pattern permutation test. The statistic was the summed resultant length of $\Delta\varphi$ across coordinates, compared with a null distribution (10,000 permutations) in which coordinate labels were shuffled within each participant. A significant result would indicate that the color–orientation phase offset varied systematically across spatial coordinates, whereas a non-significant result indicates a consistent offset that does not depend on target location.

Results

Behavioral results

Mean absolute color error was 15.7° ± 6.4° (0.27 ± 0.11 rad) in the low-load condition and 32.8° ± 18.3° (0.57 ± 0.32 rad) in the high-load condition. Mean absolute orientation error was 12.5° ± 9.7° (0.22 ± 0.17 rad) and 24.6° ± 9.5° (0.43 ± 0.17 rad) for low and high load, respectively. Response times did not differ markedly across conditions (color: 2.9 s vs. 2.6 s; orientation: 3.7 s vs. 3.5 s), with orientation reports taking consistently longer than color reports. Individual distributions of memory errors for color and orientation across participants are presented in Appendix.

Color and orientation errors showed no significant statistical dependence: mutual information did not exceed the within-participant permutation null at either low load ($p = 0.43$) or high load ($p = 0.07$). There was no significant difference in the spatial distribution of STM errors: there was no main effect of Location ($F(4,80) = 0.87$, $p = .49$) and no interaction of Location with Feature ($F(4,80) = 1.31$, $p = .27$) or Load ($F(4,80) = 1.55$, $p = .20$), indicating that memory precision was spatially uniform across the screen. As expected, memory error increased with memory load ($F(1,20) = 93.9$, $p < .001$), and was lower for color than orientation ($F(1,20) = 10.08$, $p = .005$), with this feature advantage larger at higher load (Feature × Load: $F(1,20) = 6.32$, $p = .021$).

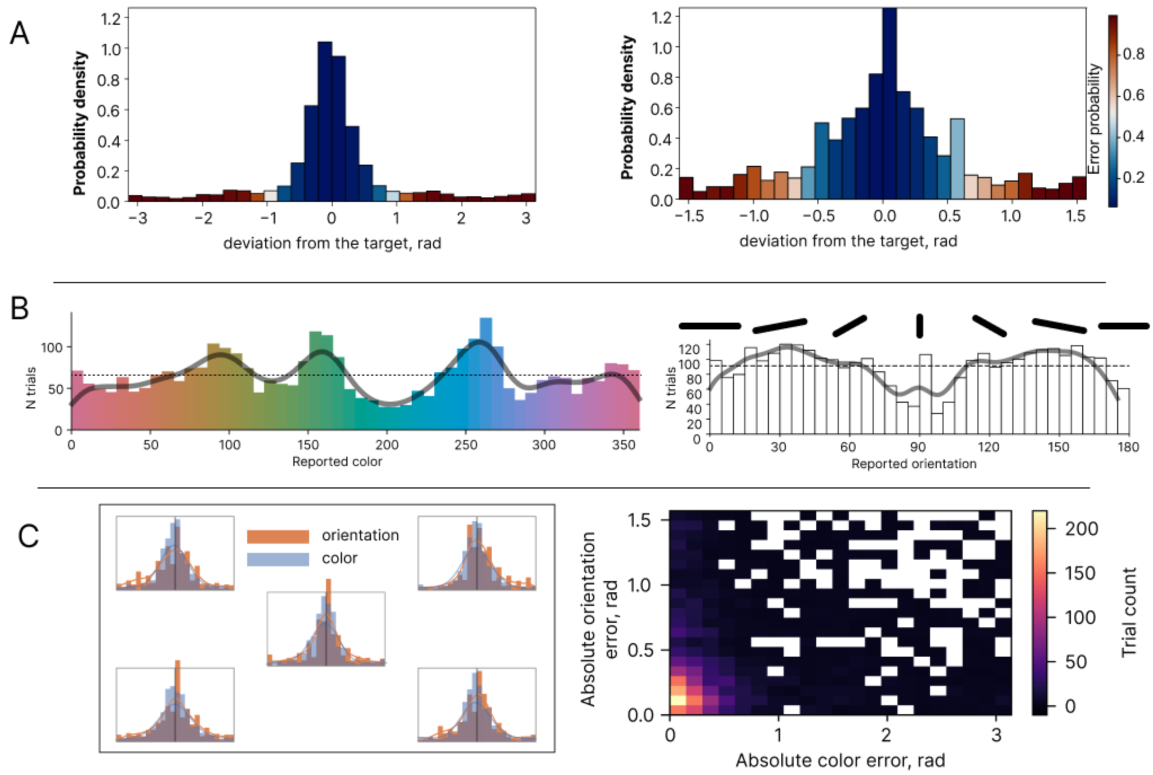

Figure 2. Behavioral STM performance across sensory features and memoranda locations.

A. Distributions of color and orientation memory errors pooled across trials. Bars are color-coded according to the fitted model-based error probability (EP)*, with higher values indicating a greater probability that the response reflected an error (forgotten) item. Orientation errors were more broadly distributed than color errors.*

B. Group-level distributions of reported colors and orientations. The non-uniform structure of both distributions suggests a preference of color and orientation values within each category.

C. Memory errors control analyses. Left: color and orientation error distributions are shown separately for different spatial positions. No reliable differences in error magnitude were observed across positions. Right: two-dimensional distribution of absolute color and orientation errors.

Time-resolved error prediction results

Time-resolved MVPA identified multiple temporal windows within the encoding period, in which neural patterns significantly predicted subsequent memory error. In both memory load conditions, prediction of color and orientation error exhibited an oscillating pattern. Figure 3A presents the group-level prediction time courses with significant clusters identified by cluster-based permutation testing; a similar pattern was observed for within-subject regression models (Appendix).

Peak frequency analysis revealed that memory error prediction for both color and orientation fluctuated at a mean frequency of 4 Hz (range: 3–5 Hz) for both low and high-load conditions (Figure 3B).

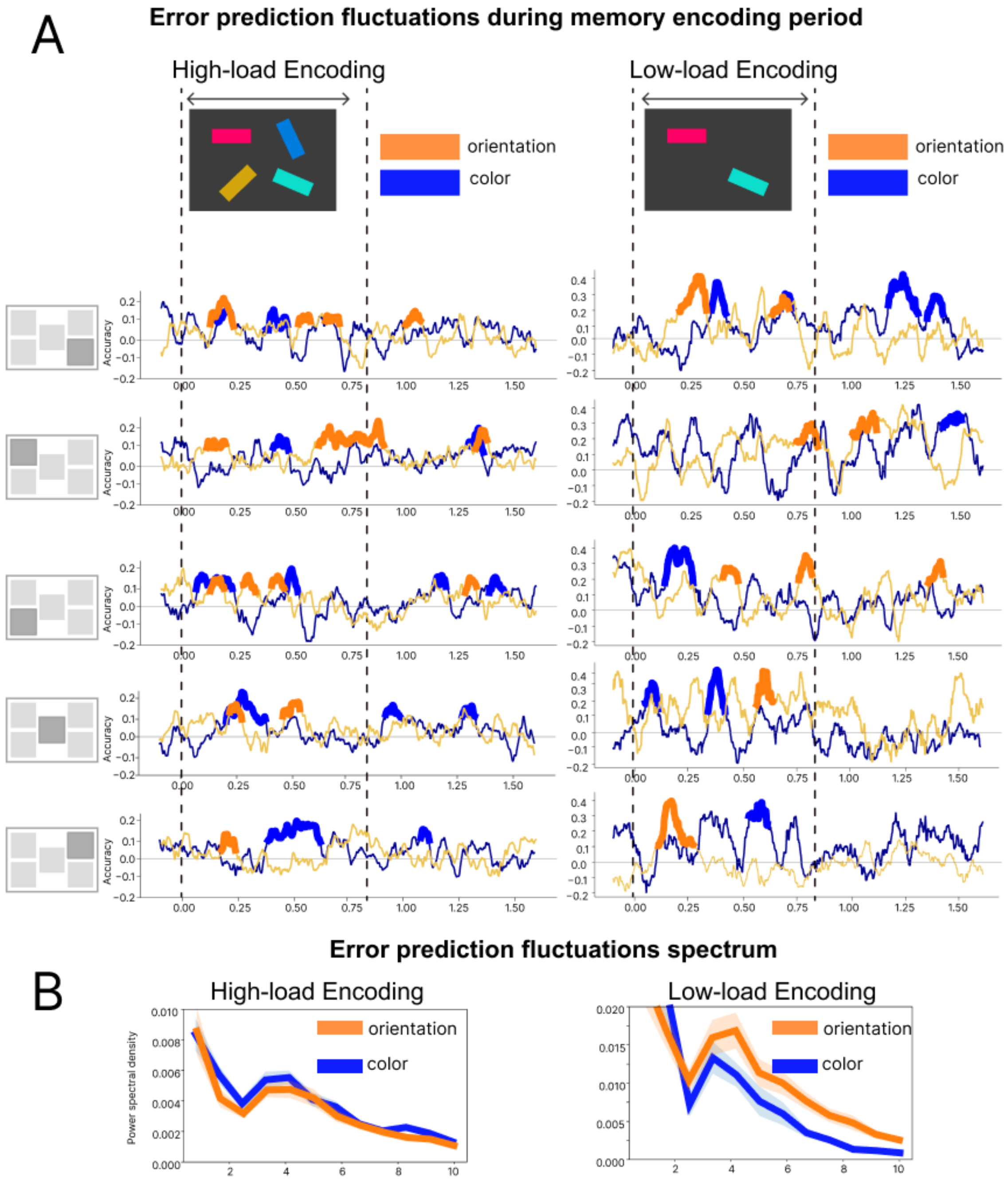

Figure 3. Time-resolved and spectral dynamics of MVPA prediction of color and orientation memory errors during encoding period.

A. Group-level time courses of errors prediction for subsequent color and orientation memory errors in the high-load and low-load memory-load conditions for different spatial positions. Prediction values are shown across the encoding period and the following maintenance interval. In both load conditions, prediction of color and orientation errors exhibited a fluctuating temporal pattern. Thickened line segments indicate significant temporal clusters identified with cluster-based permutation testing.

B. Spectral power density of the prediction fluctuations, averaged across subjects and spatial positions. For both color and orientation prediction fluctuated predominantly within the 3–5 Hz range in both memory-load conditions.

Spatial and feature-specific phase structure of the 4 Hz decoding rhythm

A Watson–Williams circular ANOVA indicated that the mean 4 Hz phase of the error-prediction decoding time courses differed across the five target locations, $F(4, 200) = 2.56$, $p = .039$.

To test for a systematic phase shift between color and orientation encoding, we estimated the phase lag between color and orientation error-prediction dynamics across all spatial positions using circular statistics at 4 Hz. In the high-load condition, color–orientation phase differences were consistently oriented across participants (subject-blocked permutation Rayleigh test: $z = 4.94$, $p = 0.036$, $r = 0.22$), clustering around a mean offset of approximately −53° (≈37 ms at 4 Hz). This offset did not differ reliably across spatial positions (within-subject condition-pattern permutation: statistic = 1.51, $p = 0.13$), indicating a consistent, location-independent phase relationship. These results indicate that orientation error prediction preceded color error prediction. No significant phase-lag effect was observed in the low-load condition.

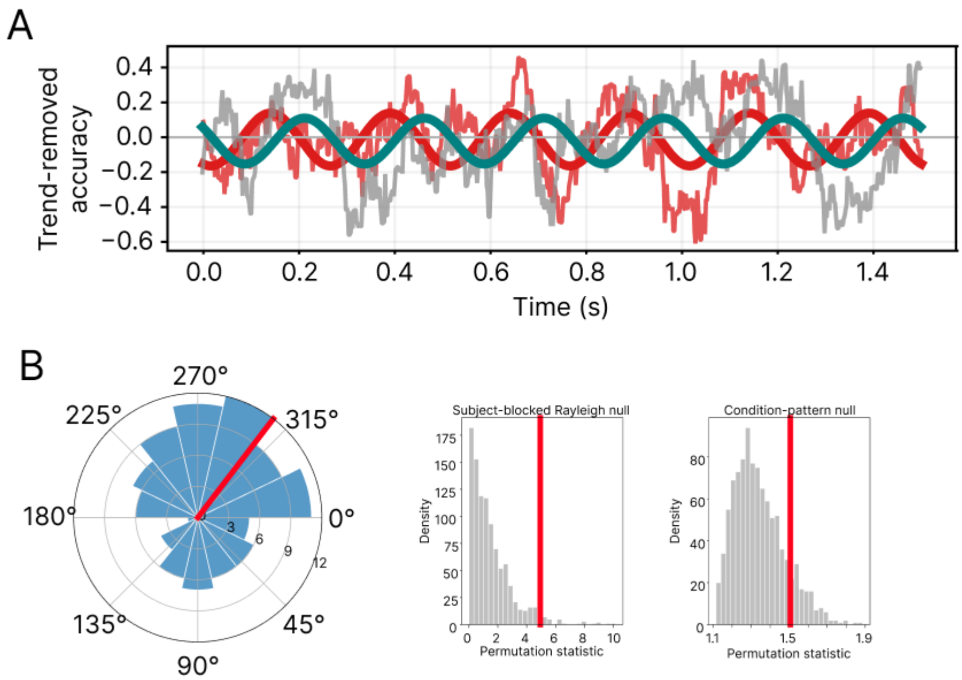

Figure 4. Orientation error prediction precedes color error prediction during encoding.

A. Representative participant showing color (grey) and orientation (red) error-prediction time courses together with fitted 4-Hz sinusoids.

B. Color–orientation phase coupling at 4 Hz. Circular histogram of color–orientation phase differences ($\Delta\varphi = \varphi\_color - \varphi\_orientation$) across participants and spatial locations; the red radial line denotes the group circular mean. Middle and right panels show permutation null distributions for the subject-blocked Rayleigh test and the within-subject condition-pattern test, respectively. Red vertical lines indicate the observed test statistics.

Cross-temporal generalization

To determine whether the fluctuating prediction dynamics reflected reactivation of similar or distinct neural patterns, we performed a cross-temporal generalization (CTG) analysis (King & Dehaene, 2014). If the error-prediction peaks observed in the time-resolved MVPA result reflect a recurring shared neural code (i.e., similar patterns), a model trained at one prediction peak should generalize to another, producing significant off-diagonal clusters at their intersection in the time × time matrix. Conversely, if each recurrence involves a distinct pattern, significant prediction should be limited to the diagonal region.

This analysis was conducted using within-subject regression models because CTG requires neural representations to generalize across time points, making it more sensitive to temporal variability than standard decoding. Within-subject models increase sensitivity by capturing participant-specific spatial and spectral response patterns.

For both color and orientation CTG revealed significant clusters off the diagonal, indicating that a shared neural pattern predicting memory error reappears several times across the encoding and maintenance period (Figure 5, A shows CTG matrix for one spatial position, for all positions see Appendix).

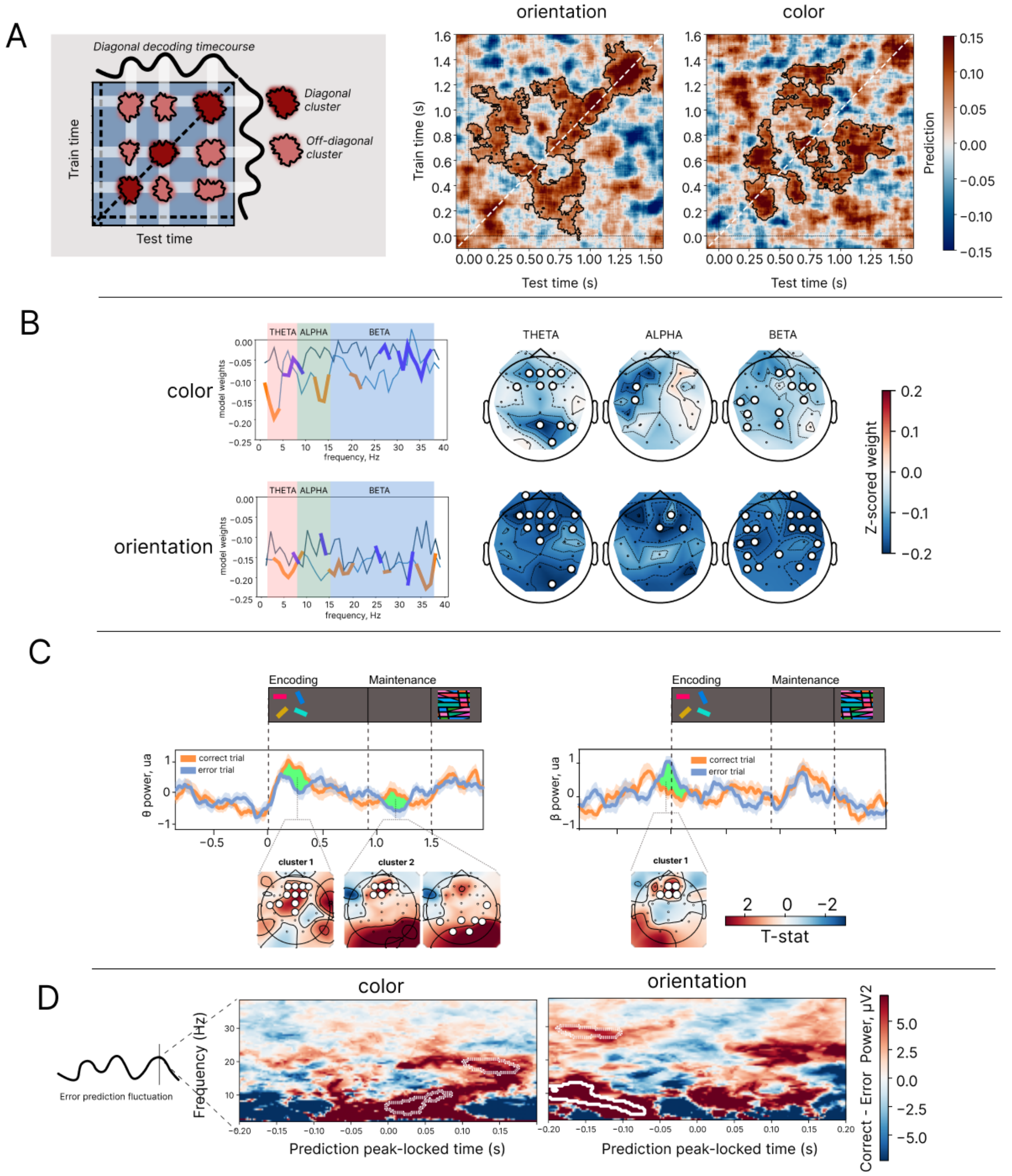

Figure 5. Neural representations underlying memory error prediction. Temporal generalization analysis shows periodic reinstatement of the same neural pattern. Analysis of the spatial and spectral characteristics of prediction-enabling neural patterns revealed dominant contributions of theta- and beta-band activity.

A. Cross-temporal generalization of error-prediction decoding. Left: schematic of the pattern expected if a shared neural representation is periodically reinstated — a rhythmic diagonal decoding time course together with recurring off-diagonal clusters in the train-time × test-time matrix. Right: empirical temporal-generalization matrices for orientation and color at a single representative target coordinate show that, in addition to diagonal clusters of significant error decoding, there are also significant off-diagonal clusters.

B. Model-weight analysis of the spectral and topographic contributions to error prediction, for color (top) and orientation (bottom). Left: model weights as a function of frequency for frontal and parieto-occipital ROIs. Bold segments indicate frequencies contributing significantly. Right: scalp distributions of model weights per band, with significant channels marked.

C. Time-resolved spectral power differences between correct and error trials in the high-load condition. Correct trials showed stronger theta power during encoding and maintenance, with significant frontal and occipital clusters. Error trials showed stronger frontal beta power before stimulus onset. Statistical significance was assessed using cluster-based permutation testing. Topographical maps show significant clusters, with color indicating the corresponding t-statistic.

D. Prediction-locked time-frequency analysis. For each participant, EEG spectral power within ±200 ms of error-prediction peaks was baseline-corrected against a surrogate distribution and contrasted between correct and error memory trials (Correct − Error power, μV^2) using cluster-based permutation. For color error two clusters emerged at trend level, for orientation two significant clusters were found.

Spectral and spatial contributions to STM error prediction

Model weight analysis. We next characterized the spatial and spectral features of error-predictive neural patterns by examining regression weights averaged across significant prediction windows. Significant contributions were distributed across theta, alpha, and beta bands and were predominantly negative, indicating that stronger oscillatory power predicted lower subsequent error (Figure 5, B). Theta-band weights showed a fronto-posterior distribution, alpha-band contributions were mainly anterior, and beta-band weights were distributed across frontal and posterior sites.

Spectral power analysis. To demonstrate what is happening in the raw oscillatory power in correct and error trials we first ran a pairwise comparison between spectral power dynamics during memory encoding in correct and error trials (only the high memory-load condition was analyzed, because it provided a sufficient number of memory errors for reliable comparisons).

To formally divide the trials into correct and error trials, we used a fitted mixture model, computing an error probability (EP) for each trial. Correct trials were defined as those for which the error probability for at least one feature was less than 0.25. Error trials were defined as those for which the EP for color or orientation memory was greater than 0.75 (see Appendix for more details).

Figure 5,C shows theta power dynamics in correct and error trials during the encoding and maintenance intervals. A cluster-based permutation test revealed a significant frontal cluster during encoding, with greater theta power in correct than in error trials ($p < 0.05$, Cohen's d = 1.13). During the maintenance phase, two significant clusters emerged, indicating stronger theta power in correct trials over frontal and occipital regions ($p < 0.05$; Cohen's d = 1.0, 1.0, correspondingly). Additionally, in the beta frequency range a significant frontal cluster was identified. Notably, this cluster emerged before stimulus onset ($p < 0.05$, Cohen's d = 1.0).

Prediction-locked spectral power. Additionally, we compared spectral perturbations in time windows around moments when error prediction peaks. For each subject EEG spectral power was analyzed within ±200 ms of error prediction peaks (for color and orientation separately). Peak-locked activity was baselined to surrogate signal and then compared between correct and error memory trials using the permutation cluster test. Full methodological details are provided in the Appendix.

For orientation errors, peak-locked time–frequency analysis identified a significant theta-band cluster ($p < 0.05$, Cohen's d = 0.69), whereas an identified beta-band cluster did not reach significance ($p = 0.10$, Cohen's d = 0.69). For color errors, no clusters survived correction, although trend-level theta and beta clusters were observed (theta: $p = 0.09$, Cohen's d = 0.78; beta: $p = 0.10$, Cohen's d = 0.79).

Discussion

Brain oscillations in the theta and beta frequency ranges have been consistently implicated in the encoding and processing of memory content, coordinating neural activity across distributed brain regions (Hanslmayr & Staudigl, 2014; Wynn et al., 2024; Scholz et al., 2017; Schmidt et al., 2019; Chang et al., 2023). Recent evidence has begun to revise the traditional view that memory functions are localized to discrete brain regions and supported by separable oscillatory frequencies. Instead, memory is increasingly understood as emerging from distributed interactions across cortical networks and oscillatory rhythms (Christophel et al., 2017; Schmidt et al., 2019; Han et al., 2026). In the present study, we provide evidence that a spatially distributed, multi-frequency neural pattern during short-term memory encoding predicts memory fidelity. This pattern does not appear as sustained activation within a fixed time window after encoding onset, but instead oscillates at a theta frequency during encoding and maintenance. This oscillatory recurrence suggests a periodic replay of memory content rather than a strictly serial process, in which each object is encoded within a single time window. We further observed a temporal offset between the prediction of multimodal object features, supporting prior proposals that the features of a multi-feature object are encoded separately rather than as a unified representation (Schneegans & Bays, 2017).

According to traditional theories of short-term memory, the temporal maintenance of encoded information depends on persistent neural activity established at encoding and sustained throughout the maintenance period (Sreenivasan et al., 2014). Recent findings challenge this account, demonstrating that mnemonic information can be retained in the absence of sustained firing (Rose et al., 2016; Stokes, 2015; Kandemir et al., 2024). Under activity-silent frameworks, memory traces are instead maintained within transient, spatially distributed neural circuits — for example, through short-term synaptic plasticity — that undergo theta-orchestrated reactivation during encoding and maintenance (Stokes, 2015; Miller et al., 2018; Pals et al., 2020).

In contrast to sustained activity, such rhythmic reinstatement enables a temporal segregation of competing memory representations, thereby reducing interference in multi-item working memory (Abdalaziz et al., 2023; Leszczyński et al., 2015; Fuentemilla et al., 2010; Lisman & Jensen, 2013).

The present findings speak directly to this framework. Using time-resolved decoding of memory errors, we found that distributed, multi-frequency neural patterns associated with STM fidelity were neither sustained nor confined to a single critical moment of memory formation. Instead, these patterns recurred rhythmically at theta frequency throughout encoding and into the maintenance period.

A conceptually similar decoding approach was previously used by Fuentemilla et al. (2010), who showed that widespread distributed occipital, frontal, and parietal activity during memory maintenance enabled differentiation of stored sensory content. Using a classification approach to predict the remembered scene category (indoor vs. outdoor), they found that classification accuracy fluctuated at theta frequency, pointing to periodic reactivation of memory content, which supports short-term retention. Similar results were found in Kerrén et al., (2018). In the present study we predicted continuous memory error, and our results converge with these reports in revealing theta-rhythmic fluctuations of decoding. It is interesting to note though that Fuentemilla et al. found that the number of replay events scaled with maintenance demands, whereas we found the reinstatement dynamics to be independent of memory load. Our work further extends Fuentemilla et al. (2010) and Kerrén et al., (2018) findings through the cross-temporal generalization analysis, which revealed that the recurring prediction reflected the same neural pattern recurring cyclically throughout encoding and into maintenance, rather than a succession of distinct neural codes.

How, then, are the different items and features of a multi-item scene or multi-feature object represented within these recurring cycles? The revealed theta-fluctuating reinstatement of STM patterns aligns with the theta-gamma phase coding model (Lisman & Idiart, 1995), in which each item is represented by a neural assembly firing at a distinct phase of the theta cycle (Jensen & Tesche, 2002; Sauseng et al., 2010; Axmacher et al., 2010; Heusser et al., 2016). Theta oscillations thus serve as a temporal scaffold for memory processes, while transient high-frequency activity in the gamma and beta bands carries and controls its content through their dynamic interplay (Lundqvist et al., 2016; Miller et al., 2018).

Consistent with such a temporal scaffold organizing the memory content, Han et al. (2026) recently reported that, in non-human primates, behavioral performance during STM retrieval, including response time and accuracy, fluctuated at theta frequency and was coupled to the phase of frontal theta oscillations (3–6 Hz). Critically, these fluctuations had different phases for different spatial locations, which the authors interpreted as a sequential theta wave "scanning" memory representations across visual space. Similarly, the theta-rhythmic STM replay patterns we observed were phase-offset across spatial locations, suggesting that spatial representations in STM may be temporally organized within the theta cycle.

Extending this spatial organization, we also found that reinstatement of STM patterns for different sensory features of the same object was also temporally offset. During encoding, color information was predicted earlier than orientation information.

This temporal dissociation of color and orientation extends the principle of rhythmic sampling to the level of features within an object. Fiebelkorn et al. (2013) showed that, even under sustained attention, perceptual sensitivity is rhythmically reweighted between objects, with the two objects sampled in antiphase at ~4 Hz. Our findings suggest that the features of a single object are likewise sampled individually in STM, each within a separate ~4 Hz process. The observed temporal separation of feature encoding extends previous STM studies showing that neural signals during encoding and maintenance can reflect temporally separated memory processing of different scene features (Cichy et al., 2017; Bocincova & Johnson, 2019; Frenkel & Deouell). It is also consistent with the behavioral independence of color and orientation reports observed in our data and provides a candidate mechanism for temporally segregating otherwise overlapping feature representations (Bays et al., 2009; Schneegans & Bays, 2017).

However, the direction of this temporal separation (orientation preceding color during encoding) contrasts with evidence from asynchronous perception, where color is typically perceived before orientation by ~50–60 ms (Moutoussis & Zeki, 1997a, 1997b). This suggests that features that appear earlier in the perceptual hierarchy might be encoded / replayed later in the memory-related theta cycle, although this hypothesis needs further testing. In addition to the main encoding-period analysis, we examined retrieval-period decoding dynamics (see Appendix) and observed an inversion of the color/orientation temporal separation, with color decoding preceding orientation decoding by approximately 72 ms.

Notably, these rhythmic reinstatements have been revealed across studies using markedly different methods. Han et al. (2026) found that memory accuracy fluctuated with theta depending on the phase at which memoranda were presented, while Ter Wal et al. (2021) demonstrated that theta rhythmicity is detectable in behavioral responses themselves, with button-press timestamps indexing the moment of memory formation rhythmically modulated at 1–5 Hz. With our MVPA prediction framework, we extend this logic in a specific way: not only does the timing of memory operations fluctuate at theta, but the quality of encoding — indexed by subsequent recall error — does so as well, and we demonstrate this directly in the recurring neural patterns. In doing so, we provide direct evidence reconciling behavioral and neural oscillations. We propose that behavioral responses can serve as a proxy for the neural clocking of memory processes: the prediction of behavioral output can reveal this clocking, too.

Spatially distributed theta and beta power contributions

Pairwise comparison of spectral power between correct and error trials revealed stronger frontal and occipital theta power on correct trials during both the encoding and maintenance periods, consistent with prior work linking theta to STM performance (Jensen & Tesche, 2002;

Sauseng et al., 2010). In contrast, frontal beta power during the encoding-onset window was stronger on trials with poor subsequent memory. This directionality is consistent with reports linking beta desynchronization — that is, beta power decreases — to successful memory formation: Hanslmayr et al. (2014) showed that prefrontal beta power decreased more strongly for subsequently remembered than forgotten items, and that artificially increasing prefrontal beta through rhythmic TMS entrainment selectively impaired encoding, establishing a causal link between elevated prefrontal beta and poorer memory. If reduced beta favors encoding, the elevated frontal beta observed here on poorly remembered trials reflects the converse, less favorable state.

Consistent with these univariate effects, analysis of the MVPA model weights showed that frontal and parietal–occipital channels in the theta and beta ranges contributed most strongly to error prediction. Together, these results indicate that STM fidelity is supported by spatially distributed, multi-frequency activity rather than by a single oscillatory signature or cortical locus (Christophel et al., 2017; Schmidt et al., 2019; Han et al., 2026).

Attention and saccadic scanning as possible explanations

Theta-rhythmic fluctuations in performance have also been described to underlie attentional processes. Spatial attention can sample multiple locations at ~4Hz periodicity (Landau & Fries, 2012; Fiebelkorn & Kastner, 2019; Re et al., 2019), with the underlying frontoparietal network alternating between exploitation and exploration states (Fiebelkorn et al., 2018; Sauseng et al., 2010). Under this account, each attentional visit to a memoranda location, e.g. each gaze fixation, would generate a new encoding episode, such that the theta-fluctuation in prediction accuracy would reflect the temporal structure of attentional sampling rather than a memory-specific process.

Several observations argue against this purely attentional explanation. First, error-prediction fluctuations were also present during the maintenance interval, when no stimulus was presented. Second, the fluctuations were not coupled to eye movements: cross-correlation between EOG velocity and decoding time courses revealed no systematic relationship for either feature or load condition (see Appendix). Third, to test whether the same rhythmic architecture operates beyond encoding, we extended the analysis to the retrieval interval (0-4s after spatial-cue onset) and found comparable ~4 Hz fluctuations, with similar phase offsets across spatial positions and across features, even though the screen showed only a position cue, with no colored or oriented items. Moreover, the temporal relationship between color and orientation reversed relative to encoding During encoding, color prediction lagged behind orientation prediction, whereas during retrieval this ordering was inverted, with color prediction preceding orientation prediction (see Appendix for details). This reversal is consistent with theories proposing that the direction of information flow differs between memory encoding and retrieval (Linde-Domingo et al., 2019; Staresina & Wimber, 2019), although this interpretation remains speculative and requires further investigation.

Nonetheless, attentional sampling and memory processing need not be competing explanations. Both mechanisms fit the embedded-process framework of Köster and Gruber (2022), in which theta oscillations serve as a unifying pacemaker for attention and memory and under this view, attentional sampling and memory encoding are not distinct operations (see Köster & Gruber, 2022).

Conclusion

Our findings provide direct evidence that working-memory encoding fidelity, indexed by subsequent recall-error prediction, fluctuates rhythmically at theta frequency during encoding and maintenance. These fluctuations were expressed as a distributed, multi-frequency neural pattern that was repeatedly reinstated across the trial. The results further suggest that items and their features are sampled at distinct phases of the theta rhythm, providing a candidate mechanism for segregating overlapping representations. Building on these findings, future work may enable closed-loop approaches that predict memory errors from ongoing neural activity and use temporally targeted stimulation to modulate memory performance.

Author Contributions

NS: Conceptualization, investigation, data curation, formal analysis, methodology, visualization, writing – original draft, writing – review & editing. **SS:** Data curation, writing – original draft. **RR:** Data curation, formal analysis. **XK:** Conceptualization, investigation, methodology, validation, resources, supervision, project administration, data curation, writing – review & editing.

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Ethics Statement

The study was approved by the local ethics committee.

Conflicts of Interest

The authors declare no conflicts of interest.

References

Abdalaziz, M., Redding, Z. V., & Fiebelkorn, I. C. (2023). Rhythmic temporal coordination of neural activity prevents representational conflict during working memory. Current Biology, 33(9), 1855–1863.e3.

Axmacher, N., Henseler, M. M., Jensen, O., Weinreich, I., Elger, C. E., & Fell, J. (2010). Cross-frequency coupling supports multi-item working memory in the human hippocampus. Proceedings of the National Academy of Sciences, 107(7), 3228–3233.

Bays, P. M., Catalao, R. F. G., & Husain, M. (2009). The precision of visual working memory is set by allocation of a shared resource. Journal of Vision, 9(10), 7, 1–11.

Berens, P. (2009). CircStat: A MATLAB toolbox for circular statistics. Journal of Statistical Software, 31(10), 1–21.

Bocincova, A., & Johnson, J. S. (2019). The time course of encoding and maintenance of task-relevant versus irrelevant object features in working memory. Cortex, 111, 196–209.

Chang, W.-S., Liang, W.-K., Li, D.-H., Muggleton, N. G., Balachandran, P., Huang, N. E., & Juan, C.-H. (2023). The association between working memory precision and the nonlinear dynamics of frontal and parieto-occipital EEG activity. Scientific Reports, 13(1), 14252.

Christophel, T. B., Klink, P. C., Spitzer, B., Roelfsema, P. R., & Haynes, J.-D. (2017). The distributed nature of working memory. Trends in Cognitive Sciences, 21(2), 111–124.

Cichy, R. M., Khosla, A., Pantazis, D., & Oliva, A. (2017). Dynamics of scene representations in the human brain revealed by magnetoencephalography and deep neural networks. NeuroImage, 153, 346–358.

Coltheart, M. (1983). Iconic memory. Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences, 302(1110), 283–294.

Fiebelkorn, I. C., & Kastner, S. (2019). A rhythmic theory of attention. Trends in Cognitive Sciences, 23(2), 87–101.

Fiebelkorn, I. C., Pinsk, M. A., & Kastner, S. (2018). A dynamic interplay within the frontoparietal network underlies rhythmic spatial attention. Neuron, 99(4), 842–853.e8.

Fiebelkorn, I. C., Saalmann, Y. B., & Kastner, S. (2013). Rhythmic sampling within and between objects despite sustained attention at a cued location. Current Biology, 23(24), 2553–2558.

Frenkel, C., & Deouell, L. Y. (n.d.). EEG-based decoding of color and visual category representations is reliable within and across sessions. [Year and source to be confirmed.]

Fuentemilla, L., Penny, W. D., Cashdollar, N., Bunzeck, N., & Düzel, E. (2010). Theta-coupled periodic replay in working memory. Current Biology, 20(7), 606–612.

Gramfort, A., Luessi, M., Larson, E., Engemann, D. A., Strohmeier, D., Brodbeck, C., Goj, R., Jas, M., Brooks, T., Parkkonen, L., & Hämäläinen, M. (2013). MEG and EEG data analysis with MNE-Python. Frontiers in Neuroscience, 7, 267.

Han, H. B., Brincat, S. L., Buschman, T. J., & Miller, E. K. (2026). Working memory readout varies with frontal theta rhythms. Neuron, 114(1), 159–166.

Hanslmayr, S., Matuschek, J., & Fellner, M.-C. (2014). Entrainment of prefrontal beta oscillations induces an endogenous echo and impairs memory formation. Current Biology, 24(8), 904–909.

Hanslmayr, S., & Staudigl, T. (2014). How brain oscillations form memories — A processing based perspective on oscillatory subsequent memory effects. NeuroImage, 85, 648–655.

Hasselmo, M. E., Bodelón, C., & Wyble, B. P. (2002). A proposed function for hippocampal theta rhythm: Separate phases of encoding and retrieval enhance reversal of prior learning. Neural Computation, 14(4), 793–817.

Heusser, A. C., Poeppel, D., Ezzyat, Y., & Davachi, L. (2016). Episodic sequence memory is supported by a theta–gamma phase code. Nature Neuroscience, 19(10), 1374–1380.

Hsieh, L.-T., & Ranganath, C. (2014). Frontal midline theta oscillations during working memory maintenance and episodic encoding and retrieval. NeuroImage, 85, 721–729.

Jas, M., Engemann, D. A., Bekhti, Y., Raimondo, F., & Gramfort, A. (2017). Autoreject: Automated artifact rejection for MEG and EEG data. NeuroImage, 159, 417–429.

Jensen, O., Gelfand, J., Kounios, J., & Lisman, J. E. (2002). Oscillations in the alpha band (9–12 Hz) increase with memory load during retention in a short-term memory task. Cerebral Cortex, 12(8), 877–882.

Jensen, O., & Tesche, C. D. (2002). Frontal theta activity in humans increases with memory load in a working memory task. European Journal of Neuroscience, 15(8), 1395–1399.

Kalcher, J., & Pfurtscheller, G. (1995). Discrimination between phase-locked and non-phase-locked event-related EEG activity. Electroencephalography and Clinical Neurophysiology, 94(5), 381–384.

Kandemir, G., Wilhelm, S. A., Axmacher, N., & Akyürek, E. G. (2024). Maintenance of color memoranda in activity-quiescent working memory states: Evidence from impulse perturbation. iScience, 27(4).

Kerrén, C., Linde-Domingo, J., Hanslmayr, S., & Wimber, M. (2018). An optimal oscillatory phase for pattern reactivation during memory retrieval. Current Biology, 28(21), 3383–3392.e6.

King, J.-R., & Dehaene, S. (2014). Characterizing the dynamics of mental representations: The temporal generalization method. Trends in Cognitive Sciences, 18(4), 203–210.

Köster, M., & Gruber, T. (2022). Rhythms of human attention and memory: An embedded process perspective. Frontiers in Human Neuroscience, 16, 905837.

Landau, A. N., & Fries, P. (2012). Attention samples stimuli rhythmically. Current Biology, 22(11), 1000–1004.

Leszczyński, M., Fell, J., & Axmacher, N. (2015). Rhythmic working memory activation in the human hippocampus. Cell Reports, 13(6), 1272–1282.

Linde-Domingo, J., Treder, M. S., Kerrén, C., & Wimber, M. (2019). Evidence that neural information flow is reversed between object perception and object reconstruction from memory. Nature Communications, 10, 179.

Lisman, J. E., & Idiart, M. A. P. (1995). Storage of 7 ± 2 short-term memories in oscillatory subcycles. Science, 267(5203), 1512–1515.

Lisman, J. E., & Jensen, O. (2013). The theta–gamma neural code. Neuron, 77(6), 1002–1016.

Lundqvist, M., Rose, J., Herman, P., Brincat, S. L., Buschman, T. J., & Miller, E. K. (2016). Gamma and beta bursts underlie working memory. Neuron, 90(1), 152–164.

Luu, P., Tucker, D. M., & Makeig, S. (2004). Frontal midline theta and the error-related negativity: Neurophysiological mechanisms of action regulation. Clinical Neurophysiology, 115(8), 1821–1835.

Miller, E. K., Lundqvist, M., & Bastos, A. M. (2018). Working Memory 2.0. Neuron, 100(2), 463–475.

Moutoussis, K., & Zeki, S. (1997a). A direct demonstration of perceptual asynchrony in vision. Proceedings of the Royal Society B: Biological Sciences, 264(1380), 393–399.

Moutoussis, K., & Zeki, S. (1997b). Functional segregation and temporal hierarchy of the visual perceptive systems. Proceedings of the Royal Society B: Biological Sciences, 264(1387), 1407–1414.

Pals, M., Stewart, T. C., Akyürek, E. G., & Borst, J. P. (2020). A functional spiking-neuron model of activity-silent working memory in humans based on calcium-mediated short-term synaptic plasticity. PLOS Computational Biology, 16(6), e1007936.

Peirce, J. W. (2007). PsychoPy—Psychophysics software in Python. Journal of Neuroscience Methods, 162(1–2), 8–13.

Re, D., Inbar, M., Richter, C. G., & Landau, A. N. (2019). Feature-based attention samples stimuli rhythmically. Current Biology, 29(4), 693–699.e4.

Rose, N. S., LaRocque, J. J., Riggall, A. C., Gosseries, O., Starrett, M. J., Meyering, E. E., & Postle, B. R. (2016). Reactivation of latent working memories with transcranial magnetic stimulation. Science, 354(6316), 1136–1139.

Sauseng, P., Griesmayr, B., Freunberger, R., & Klimesch, W. (2010). Control mechanisms in working memory: A possible function of EEG theta oscillations. Neuroscience & Biobehavioral Reviews, 34(7), 1015–1022.

Schmidt, R., Herrojo Ruiz, M., Kilavik, B. E., Lundqvist, M., Starr, P. A., & Aron, A. R. (2019). Beta oscillations in working memory, executive control of movement and thought, and sensorimotor function. The Journal of Neuroscience, 39(42), 8231–8238.

Schneegans, S., & Bays, P. M. (2017). Neural architecture for feature binding in visual working memory. The Journal of Neuroscience, 37(14), 3913–3925.

Scholz, S., Schneider, S. L., & Rose, M. (2017). Differential effects of ongoing EEG beta and theta power on memory formation. PLOS ONE, 12(2), e0171913.

Sederberg, P. B., Kahana, M. J., Howard, M. W., Donner, E. J., & Madsen, J. R. (2003). Theta and gamma oscillations during encoding predict subsequent recall. The Journal of Neuroscience, 23(34), 10809–10814.

Sreenivasan, K. K., Curtis, C. E., & D'Esposito, M. (2014). Revisiting the role of persistent neural activity during working memory. Trends in Cognitive Sciences, 18(2), 82–89.

Staresina, B. P., & Wimber, M. (2019). A neural chronometry of memory recall. Trends in Cognitive Sciences, 23(12), 1071–1085.

Stokes, M. G. (2015). ‘Activity-silent’ working memory in prefrontal cortex: A dynamic coding framework. Trends in Cognitive Sciences, 19(7), 394–405.

ter Wal, M., Linde-Domingo, J., Lifanov, J., Roux, F., Kolibius, L. D., Gollwitzer, S., Hamer, J., Rollings, D., Sawlani, V., Chelvarajah, R., Staresina, B., Hanslmayr, S., & Wimber, M.

(2021). Theta rhythmicity governs human behavior and hippocampal signals during memory-dependent tasks. Nature Communications, 12, 7048.

Trujillo, L. T., & Allen, J. J. B. (2007). Theta EEG dynamics of the error-related negativity. Clinical Neurophysiology, 118(3), 645–668.

Wynn, S. C., Townsend, C. D., & Nyhus, E. (2024). The role of theta and gamma oscillations in item memory, source memory, and memory confidence. Psychophysiology, 61(9), e14602.

Zhou, C., Lorist, M. M., & Mathôt, S. (2022). Categorical bias as a crucial parameter in visual working memory: The effect of memory load and retention interval. Cortex, 154, 311–321.